

Chemical Bonding Unit Test · Practice Test 1

Chemistry · bond types, drawing, properties, naming, designing materials, molecular geometry and polarity

Name: _____

Date: _____

Period: _____

Time: 50 minutes. Total: 80 points. Part A: 17 multiple-choice questions, 2 points each (34). Part B: 5 free-response questions (46). Show your work in Part B. Drawings must show **every** lone pair, and brackets and charges on ions.

Reference. Electronegativity: H 2.1 · C 2.5 · N 3.0 · O 3.5 · F 4.0 · Na 0.9 · Mg 1.2 · Al 1.5 · Si 1.8 · P 2.1 · S 2.5 · Cl 3.0 · K 0.8 · Ca 1.0 · Br 2.8. Bond type from ΔEN : below 0.5 nonpolar covalent · 0.5 to 1.7 polar covalent · above 1.7 ionic. A periodic table may be used.

Part A · Multiple Choice (2 points each; circle one answer)

- Which compound contains ionic bonds?
A CO_2 **B** CaCl_2
C NH_3 **D** CH_4
- In a metallic bond, the valence electrons are
A transferred from metal atoms to non-metal atoms
B shared in pairs between two atoms
C held as lone pairs on each atom
D delocalized and free to move among positive ions
- The electronegativity difference in an H-Cl bond is 0.9. The bond is
A polar covalent **B** nonpolar covalent
C ionic **D** metallic
- How many valence electrons are in a sulfate ion, SO_4^{2-} ?
A 26 **B** 30
C 32 **D** 34
- Which describes the Lewis structure of N_2 ?
A a double bond, with two lone pairs on each N
B a triple bond, with one lone pair on each N
C a triple bond, with no lone pairs
D a single bond, with three lone pairs on each N
- Which is the correct dot diagram for potassium oxide, K_2O ?
A K-O-K, with shared pairs between the atoms
B one $[\text{K}]^{2+}$ ion and one $[\text{O}]^{2-}$ ion
C two $[\text{K}]^-$ ions and one $[\text{O}]^{2+}$ ion
D two $[\text{K}]^+$ ions and one $[\text{O}]^{2-}$ ion with eight dots
- Which substance conducts electricity when molten but not as a solid?
A copper **B** sugar
C sodium chloride **D** diamond

8. Metals can be hammered into sheets because
- A layers of ions slide past each other while the delocalized electrons keep holding them
 - B metallic bonds are weak
 - C ions with the same charge repel each other
 - D their covalent bonds can bend
9. Which substance has the lowest melting point?
- A MgO
 - B SiO₂
 - C Fe
 - D CH₄
10. What is the name of Fe₂O₃?
- A iron oxide
 - B iron(II) oxide
 - C iron(III) oxide
 - D diiron trioxide
11. What is the formula of calcium nitrate?
- A Ca(NO₃)₂
 - B CaNO₃
 - C Ca₂NO₃
 - D Ca₃N₂
12. What is the name of N₂O₄?
- A nitrogen oxide
 - B dinitrogen tetroxide
 - C nitrogen(IV) oxide
 - D dinitrogen quadroxide
13. Which type of material is best for the flexible insulating cover of a power cable?
- A a metal
 - B an alloy
 - C an ionic solid
 - D a molecular covalent material such as a plastic
14. What is the molecular geometry of NH₃?
- A trigonal planar
 - B tetrahedral
 - C trigonal pyramidal
 - D bent
15. What is the bond angle in CO₂?
- A 180°
 - B 120°
 - C 109.5°
 - D 104.5°
16. Which molecule is nonpolar?
- A H₂O
 - B CCl₄
 - C NH₃
 - D HCl
17. Why is a water molecule polar?
- A Its O-H bonds are nonpolar.
 - B It is linear.
 - C Its bent shape means the O-H bond dipoles do not cancel.
 - D It contains ions.

Part B · Free Response (show your work)

18. Bond types and properties (8 points)

(a) Complete the table. (4 points)

Substance	Type of bonding	Conducts electricity as a solid? (yes / no)	Melting point (low / high / very high)
potassium bromide, KBr			
copper, Cu			
candle wax, C ₂₅ H ₅₂			
diamond, C			

(b) Solid KBr does not conduct electricity, but KBr dissolved in water does. Explain why. (2 points)

(c) The covalent bonds inside a wax molecule are strong, yet wax melts at a low temperature. Explain why. (2 points)

19. Names and formulas (10 points)

(a) Name each compound. (5 points)

Formula	Name
MgBr ₂	
CuCl ₂	
N ₂ O ₃	
Na ₂ CO ₃	
PCl ₅	

(b) Write the formula of each compound. (4 points)

Name	Formula
aluminum sulfide	
iron(III) nitrate	
carbon tetrachloride	
ammonium phosphate	

(c) Explain why the name of CuCl_2 needs a Roman numeral but the name of MgBr_2 does not. (1 point)

20. Drawing bonds (10 points)

(a) Draw a dot diagram for the ionic compound magnesium fluoride, MgF_2 . Show the ions in brackets with their charges. (3 points)

(b) Count the valence electrons in formaldehyde, CH_2O , and draw its Lewis structure. (3 points)

(c) Count the valence electrons in the carbonate ion, CO_3^{2-} , and draw one Lewis structure for it. State how many resonance structures the ion has. (4 points)

21. Molecular geometry and polarity (14 points: 7 for each row)

Complete every column for each molecule. In the 3-D redraw, use plain lines, wedges and dashes where they are needed, and show any lone pairs on the central atom.

Formula	Valence e ⁻	Lewis structure	Electron geometry	Molecular geometry & bond angle	3-D redraw	Bond polarity (show ΔEN)	Molecular polarity
NF ₃							
CH ₂ O							

22. Designing materials (4 points)

An engineer is choosing materials for an electric kettle.

(a) The heating plate at the bottom must pass heat to the water quickly and must not crack. Which type of bonding should the material have? Give two properties that make it suitable. (2 points)

(b) The handle is made of plastic. Which type of bonding does a plastic have, and why is it a good choice for a handle? (1 point)

(c) A ceramic (ionic) disc inside the base insulates the electrical parts. Give one property that makes a ceramic suitable there, and one drawback of ceramics. (1 point)
